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Automation and Control Laboratory Report
Group: Game of Poles

Riccardo Rossini, Siddhant Samarth, Augusto Torlonia, Y.Utku Korutürk

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1 Introduction

1.1 System description

Quanser's QUBE servo 2 consists of a driven arm that rotates in the horizontal plane and a pendulum attached to the arm that is free to rotate in the vertical plane.

The system is equipped with two rotary incremental encoders, measuring the angular positions of the arm and the pendulum, and a DC motor that directly drives the horizontal arm.

The aim of the laboratory experience was to develop and validate a full system model, based on which control strategies could be designed and tuned on the true system. Ultimately, the goal is to stabilize the pendulum in the unstable equilibrium starting from the stable one.

1.2 System parameters

The relevant parameters provided by the manufacturer are:

Symbol	Description	Value
R_m	Terminal resistance	8.4Ω
k_t	Torque constant	0.042 Nm/A
k_m	Motor back-emf constant	0.042 V/(rad/s)
J_m	Rotor inertia	$4.0 \times 10^{-6} \text{ kgm}^2$
L_m	Rotor inductance	1.16 mH
m_h	Module attachment mass	0.0106 kg
J_h	Module attachment moment of inertia	$0.6 \times 10^{-6} \text{ kgm}^2$
m_r	Rotary arm mass	0.095 kg
L_r	Rotary arm length	0.085 m
m_p	Pendulum mass	0.024 kg
L_p	Pendulum length	0.129 m
	Encoder resolution in quadrature	0.176 deg/count

2 Modelling of the system

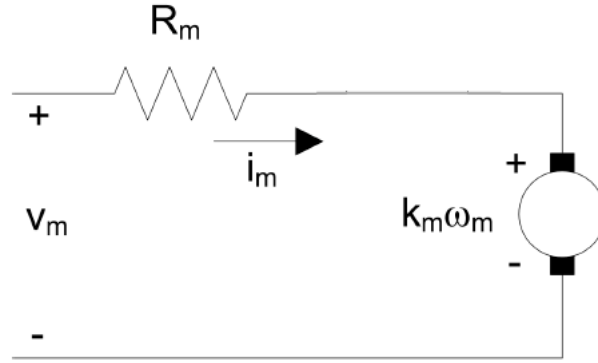
2.1 Hypothesis

Before deriving the system dynamics, some simplifying assumptions were made:

- The current dynamics were neglected because the inductance of the dc motor is several orders of magnitude smaller than the resistance. Moreover, it caused numerical instabilities with the quarc/simulink interface.
- The motor shaft and horizontal arm were assumed to be rigidly coupled and infinitely stiff.
- The pendulum was assumed to be infinitely stiff and rigidly coupled with the horizontal arm.
- The moments of inertia of the motor and the hub were considered only for the identification of the motor friction coefficient because they are several orders of magnitude smaller than those of the pendulum and arm.
- The only form of damping considered was viscous one, effectively neglecting the non linear behaviour of the static friction, even though an experiment to properly understand its magnitude has been performed.
- The encoder cable has been modeled as a symmetric torsional spring, neglecting the damping effect that it might have had on the system and its asymmetric behaviour.

2.2 Actuator model

The system actuation unit is a direct drive dc motor, whose circuit schematic (simplified following our assumptions) is:



Using Kirchoff's Voltage Law, we wrote the following equation:

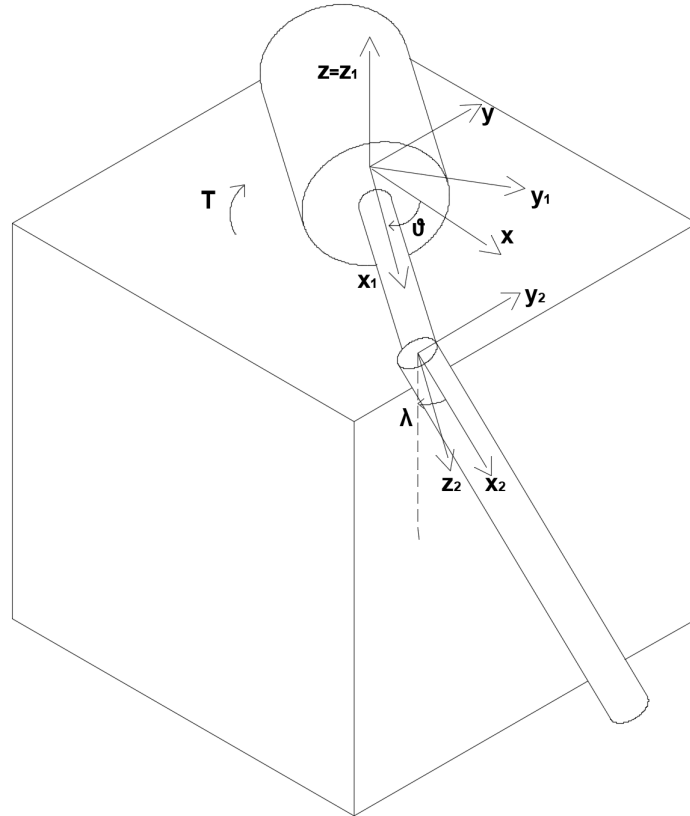
$$v_m(t) = R_m i_m(t) + k_m w_m(t)$$

Solving for the current and using the current-torque relationship $\tau = k_t i_m(t)$, we derived the torque expression:

$$T(t) = \frac{k_t(v_m(t) - k_m w_m(t))}{R_m}$$

2.3 Horizontal arm + pendulum model

The mechanical system, composed by the horizontal arm and the pendulum link, can be illustrated schematically as follows:



where:

- reference frame xyz is fixed with respect to the base
- frame 1, namely, $x_1y_1z_1$ is rigidly attached to the horizontal arm
- frame 2, namely, $x_2y_2z_2$ is rigidly attached to the pendulum link
- all frames are chosen according to the right-hand positive convention
- the horizontal arm angle θ is considered positive in the clockwise direction, so that it is coherent with the encoder readings
- the pendulum angle λ is considered positive in the counterclockwise direction, so that it is coherent with the encoder readings
- the torque is considered positive when applied in the clockwise direction, so that it is coherent with the direction of rotation corresponding to a positive input voltage being fed to the system

In order to model the system, we used the Euler-Lagrange approach, starting from the Euler-Lagrange equation

$$\frac{\partial^2 L}{\partial t \partial q} + C\dot{q} - \frac{\partial L}{\partial q} = Q^{nc},$$

with L is defined as the Lagrangian $L = K - V$; to determine the kinetic and potential energies we defined

$$\begin{aligned} w_1 &= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} : \text{angular velocity of frame 1 w.r.t. } xyz \\ v_{cm1} &= -w_1 \times \begin{bmatrix} L_{cm1} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\dot{\theta}L_{cm1} \\ 0 \end{bmatrix} : \text{velocity of the center of mass of frame 1 w.r.t. } xyz \\ w_2 &= R_2 w_1 + \begin{bmatrix} 0 \\ 0 \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} \cos(\lambda) \\ -\sin(\lambda) \\ \dot{\lambda} \end{bmatrix} \quad \text{with} \quad R_2 = \begin{bmatrix} 0 & \sin(\lambda) & -\cos(\lambda) \\ 0 & \cos(\lambda) & \sin(\lambda) \\ 1 & 0 & 0 \end{bmatrix} : \text{angular velocity of frame 2 w.r.t. } xyz \\ v_{p,frame1} &= -w_1 \times \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\dot{\theta}L_1 \\ 0 \end{bmatrix} \quad \text{and} \quad v_{p,frame2} = R_2 v_{p,frame1} = \begin{bmatrix} -\dot{\theta}L_1 \sin \lambda \\ -\dot{\theta}L_1 \cos \lambda \\ 0 \end{bmatrix} : \text{velocity of link point P} \\ v_{cm2} &= v_{p,frame2} + w_2 \times \begin{bmatrix} L_{cm2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -\dot{\theta}L_1 \sin \lambda \\ -\dot{\theta}L_1 \cos \lambda + \dot{\lambda}L_{cm2} \\ L_{cm2} \sin \lambda \dot{\theta} \end{bmatrix} : \text{velocity of center of mass of frame 2 w.r.t. } xyz \end{aligned}$$

where L_{cm1} and L_{cm2} are the center of mass lengths for the arm and pendulum, respectively, considered as half the length provided by the manufacturer.

At this point we were able to define the kinetic energy:

$$K = K_1 + K_2 = \frac{1}{2}v_{cm1}^T m_1 v_{cm1} + \frac{1}{2}w_1^T J_1 w_1 + \frac{1}{2}v_{cm2}^T m_2 v_{cm2} + \frac{1}{2}w_2^T J_2 w_2$$

Then, the potential energy (including both gravitational and elastic terms) was defined as follows:

$$V = V_g + V_k = m_2 g L_{cm2} (1 - \cos \lambda) + \frac{1}{2}k\theta^2$$

$$\text{where } J_1 = \begin{bmatrix} J_{1xx} & 0 & 0 \\ 0 & J_{1yy} & 0 \\ 0 & 0 & J_{1zz} \end{bmatrix} \quad \text{and} \quad J_2 = \begin{bmatrix} J_{2xx} & 0 & 0 \\ 0 & J_{2yy} & 0 \\ 0 & 0 & J_{2zz} \end{bmatrix} \text{ are the inertia matrices.}$$

Note that these matrices are in diagonal form owing to the choice of the axis of frames 1 and 2, which correspond to the principal axes of the arm and pendulum, respectively.

Subsequently, further simplifying assumptions regarding the arm and pendulum were made:

- Slender beam:

$$J_{1xx} = J_{2xx} = 0$$

since the longitudinal dimension is far greater than the cylindrical diameter.

- Symmetry:

$$J_{1yy} = J_{1zz} = J_1 \quad \text{and} \quad J_{2yy} = J_{2zz} = J_2$$

such that the moments of inertia in the two principal axes are equal and correspond to the moment of inertia with respect to the center of mass.

After the necessary calculation steps (omitted here for the sake of brevity), we obtained the following two non linear equations of motion:

$$\begin{cases} \ddot{\theta}(L_{cm1}^2 m_1 + m_2 L_1^2 + J_1 + L_{cm2}^2 m_2 \sin^2 \lambda + J_2 \sin^2 \lambda) - \ddot{\lambda} L_1 L_{cm2} m_2 \cos \lambda + \dot{\lambda}^2 L_1 L_{cm2} m_2 \sin \lambda + \\ + \dot{\theta} \dot{\lambda} (L_{cm2}^2 m_2 \sin 2\lambda + J_2 \sin 2\lambda) + k\theta = T - C_1 \dot{\theta} \\ \ddot{\lambda} (L_{cm2}^2 m_2 + J_2) - \ddot{\theta} (L_1 L_{cm2} m_2 \cos \lambda) + \dot{\theta}^2 (-\frac{1}{2} L_{cm2}^2 \sin 2\lambda m_2 - \frac{1}{2} J_2 \sin 2\lambda) + m_2 g L_{cm2} \sin \lambda = -C_2 \dot{\lambda} \end{cases}$$

where C_1 and C_2 are the friction coefficients for the arm and pendulum, respectively.

Therefore, the system is a 4th order, non linear, time invariant system, characterized by the following state vector:

$$\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta} \\ \lambda \\ \dot{\lambda} \end{bmatrix}$$

In order to derive a linearized model, we evaluated the system equations when all the derivatives and input are null:

$$\begin{cases} k\theta = 0 \\ m_2 g L_{cm2} \sin \lambda = 0 \end{cases}$$

which allowed us to identify the following two equilibrium points:

$(\theta = 0, \dot{\theta} = 0, \lambda = 0, \dot{\lambda} = 0)$, $(\theta = 0, \dot{\theta} = 0, \lambda = \pi, \dot{\lambda} = 0)$.

Intuitively, neglecting the static friction led us to find analytically only two equilibria. However, based on our firsthand experience with the system, when moving the pendulum manually, the elastic restorative force owing to the cable stiffness wasn't able to overcome the static friction within a delta of values for the arm angle in the neighbourhood of the 0 position, effectively increasing the number of equilibria of the system compared with the analytical ones.

At this point, we proceeded with writing the linearized system equations and deriving a suitable state space model for both equilibria:

- **Linearized model around $(\theta = 0, \dot{\theta} = 0, \lambda = 0, \dot{\lambda} = 0)$**

$$\begin{cases} \ddot{\theta} a - \ddot{\lambda} b + k\theta = T - C_1 \dot{\theta} \\ \ddot{\lambda} c - \ddot{\theta} b + \lambda d = -C_2 \dot{\lambda} \end{cases} \quad \text{where} \quad \begin{cases} a = L_{cm1}^2 m_1 + m_2 L_1^2 + J_1 \\ b = L_1 L_{cm2} m_2 \\ c = L_{cm2}^2 m_2 + J_2 \\ d = m_2 g L_{cm2} \end{cases}$$

solving for $\ddot{\theta}$ and $\ddot{\lambda}$, the system can be written in the following state space form:

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{\lambda} \\ \ddot{\lambda} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{kc}{ac-b^2} & -\frac{C_1 c}{ac-b^2} & -\frac{db}{ac-b^2} & -\frac{bC_2}{ac-b^2} \\ 0 & 0 & 0 & 1 \\ -\frac{kb}{ac-b^2} & -\frac{C_1 b}{ac-b^2} & -\frac{b^2 d}{c(ac-b^2)} - \frac{d}{c} & -\frac{b^2 C_2}{c(ac-b^2)} - \frac{C_2}{c} \end{bmatrix} \cdot \begin{bmatrix} \theta \\ \dot{\theta} \\ \lambda \\ \dot{\lambda} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{c}{ac-b^2} \\ 0 \\ \frac{b}{ac-b^2} \end{bmatrix} \cdot T$$

- **Linearized model around** ($\theta = 0, \dot{\theta} = 0, \lambda = \pi, \dot{\lambda} = 0$)

$$\begin{cases} \ddot{\theta}a + \ddot{\lambda}b + k\theta = T - C_1\dot{\theta} \\ \ddot{\lambda}c + \ddot{\theta}b - \lambda d = -C_2\dot{\lambda} \end{cases} \quad \text{where} \quad \begin{cases} a = L_{cm1}^2 m_1 + m_2 L_1^2 + J_1 \\ b = L_1 L_{cm2} m_2 \\ c = L_{cm2}^2 m_2 + J_2 \\ d = m_2 g L_{cm2} \end{cases}$$

solving for $\ddot{\theta}$ and $\ddot{\lambda}$, the system can be written in the following state space form:

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{\lambda} \\ \ddot{\lambda} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{kc}{ac-b^2} & -\frac{C_1c}{ac-b^2} & -\frac{db}{ac-b^2} & \frac{bC_2}{ac-b^2} \\ 0 & 0 & 0 & 1 \\ \frac{kb}{ac-b^2} & \frac{C_1b}{ac-b^2} & \frac{b^2d}{c(ac-b^2)} + \frac{d}{c} & -\frac{b^2C_2}{c(ac-b^2)} - \frac{C_2}{c} \end{bmatrix} \cdot \begin{bmatrix} \theta \\ \dot{\theta} \\ \lambda \\ \dot{\lambda} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{c}{ac-b^2} \\ 0 \\ -\frac{b}{ac-b^2} \end{bmatrix} \cdot T$$

3 System identification and validation

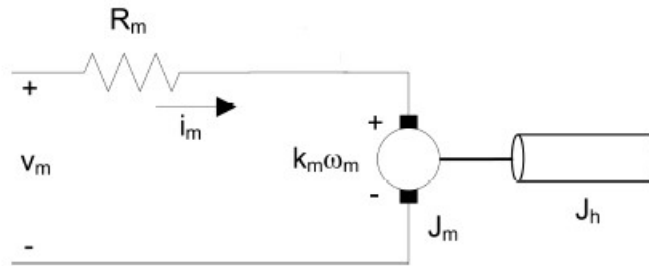
3.1 Horizontal arm friction coefficient identification

First, we conducted preliminary experiments to verify that the sensor readings were coherent with the manufacturer resolution specifications: by manually rotating both the pendulum and the arm, we checked the encoder measurements, with the angles being converted from counts to radians.

After ensuring that the readings were coherent, we physically removed the pendulum module and added a derivation block in the scheme to obtain velocity measurements. By feeding the system with a step voltage, we identified that the arm velocity measurement had high frequency components owing to the derivation being performed on a discrete signal and simulink solver using a discrete-step.

In order to solve this issue, we needed more insights on the system dynamics, so that we could properly define a low pass filter with sufficiently high cutoff frequency that would not influence the system dynamics, simultaneously ensuring that the cutoff was sufficiently low to eliminate majority of the disturbance.

The system without the pendulum module can be described schematically as follows



where J_h and J_m are the inertias of the attachment module and motor, respectively. Subsequently, the equation of motion is

$$(J_h + J_m)\ddot{\theta}(t) + C_1\dot{\theta}(t) = T(t) \quad \text{with} \quad T(t) = \frac{k_t(v_m(t) - k_m\dot{\theta}(t))}{R_m}$$

Among the equation parameters, the only one not being provided by the manufacturer is the friction coefficient. Our initial approach was to "trust" the provided parameters and verify the model accuracy, eventually identifying them only if not satisfactory.

By evaluating the equation of motion at steady state ($\ddot{\theta} = 0$), we obtained the following relationship

$$C_1 = \frac{k_t(v_m(t) - k_m\dot{\theta}(t))}{\dot{\theta}R_m}$$

Using the step response of the system to inputs in the 1-3V range, we determined

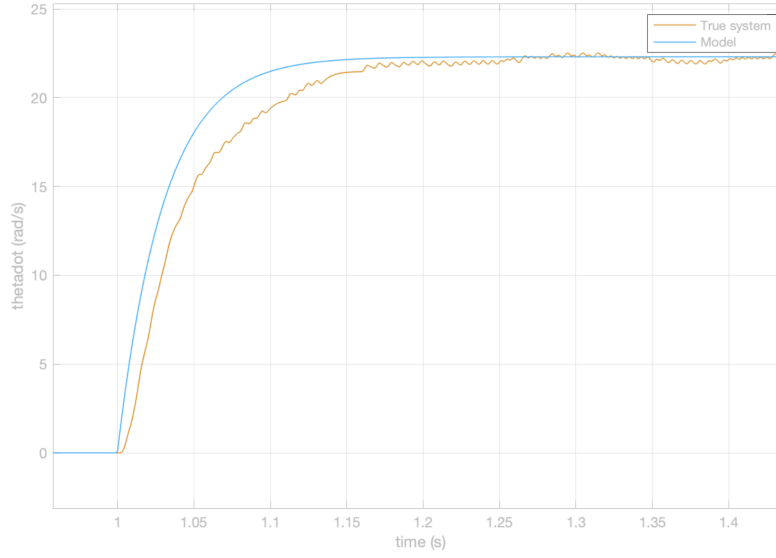
$$C_1 = 4.6 \times 10^{-4}$$

At this point, writing down the system in state space from

$$\ddot{\theta}(t) = -\frac{k_t k_m + C_1 R_m}{(J_h + J_m) R_m} \dot{\theta}(t) + \frac{k_t}{(J_h + J_m) R_m} v(t)$$

and evaluating it with respect to the parameters, we were able to properly select the cutoff frequency: since the pole of the system is at $\approx -45 \text{ rad/s}$, we chose 25Hz, approximately 5 times faster.

Comparing the model to the system we got the following result:



with an RMSE of 0.6935 rad/s.

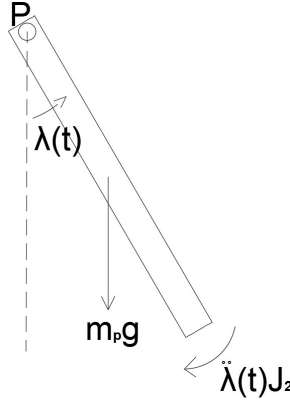
The difference is due mainly to the inertia, which will be neglected anyways, and to the non linear effect of the static friction, which introduces a small delay at the beginning. This result confirmed that our choice of trusting the parameters was reasonable.

Furthermore, we also performed a quick experiment where we manually increased the voltage, starting from very low values, until the system (with the pendulum mounted) started moving, so that we had an idea of the magnitude of the static friction. We observed motion at approximately 0.3/0.4 V.

3.2 Pendulum friction coefficient identification

For what concerns the pendulum, we performed a free response to initial conditions by manually bringing it up and letting it oscillate freely, while keeping the horizontal arm fixed. The scope of this experiment was to get an estimate of the friction coefficient while also verifying the natural frequency of oscillation and comparing it to the analytically obtained value.

The system can be schematized as follows:



and modelled as

$$\ddot{\lambda}(t)J_2 + C_2\dot{\lambda}(t) + m_2g \sin \lambda(t) = T = 0$$

where J_2 is the moment of inertia of the pendulum with respect to point P. By linearizing the model around 0 (small oscillations of the pendulum), the equation becomes:

$$\ddot{\lambda}(t)J_2 + C_2\dot{\lambda}(t) + m_2g\lambda(t) = T = 0$$

Taking as a reference the characteristic equation of second order systems, $\ddot{\lambda} + 2\xi w_0\dot{\lambda} + w_0^2\lambda = 0$, and rewriting the equation of motion to isolate $\ddot{\lambda}$, the following relationships can be derived

$$w_0 = \sqrt{\frac{mgL_{cm2}}{J_2}} = 10.68 \text{ rad/s} \quad \text{natural frequency of the pendulum}$$

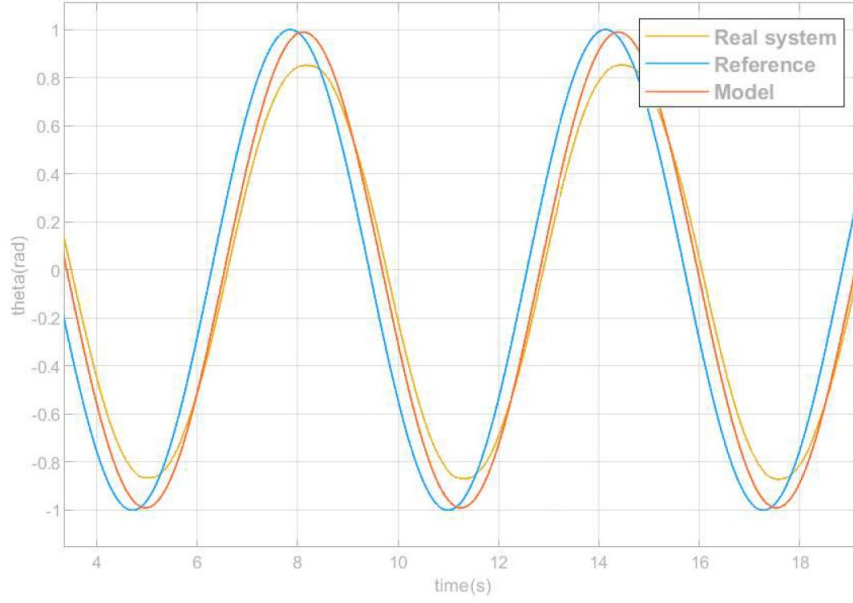
$$\xi = \sqrt{1 - \frac{w_1^2}{w_0^2}} \quad \text{damping ratio, where } w_1 \text{ is the damped frequency of oscillation we detected}$$

therefore, the friction coefficient is determined as $C_1 = 2\xi w_0 J_2$.

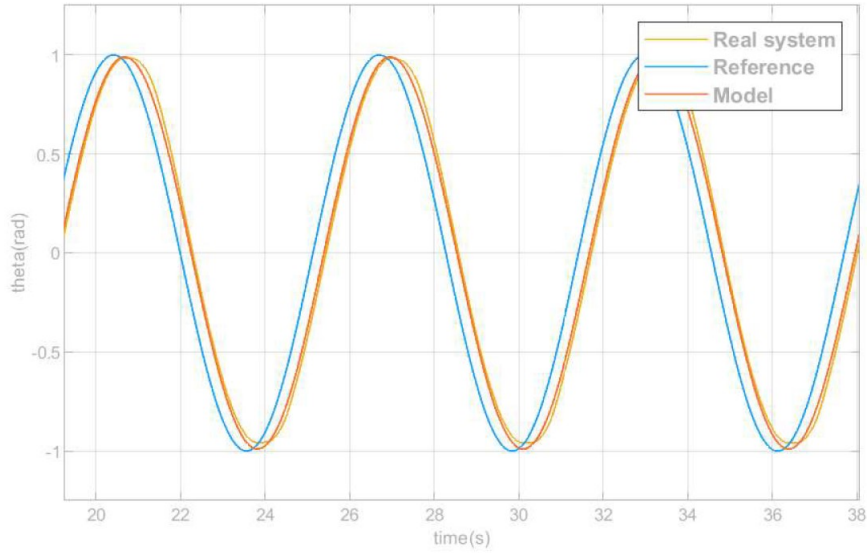
From the test we performed, the damped frequency of the system within the validity limits of the linearized model (small oscillations around 0) was on average very close to the analytical one, with cases certain where it was slightly faster and cases where it was slower. Accordingly, we decided to neglect the friction of the pendulum and consider the analytical value for the inertia of the pendulum, which led to a resonant frequency of 10.68 rad/s.

3.3 Cable stiffness identification

The influence of the cable onto the system emerged when we were working on the PID controller for the horizontal arm; the closed loop system response wasn't coherent with our model, showing a fixed offset from the reference signal



To verify that the cause of this discrepancy was indeed the cable, we unplugged and taped the encoder cable to the horizontal arm and fed the system with the same sinusoidal reference:

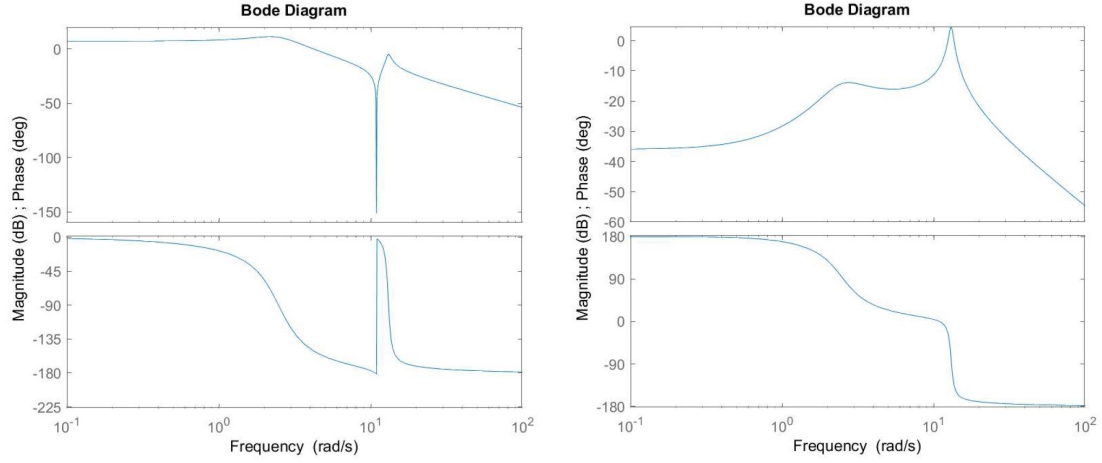


At this point, after clearly identifying the issue, we went back to modelling, representing the cable as a symmetric and linear torsional spring. Subsequently, we identified the stiffness by matching the model curve with the system one, obtaining $k = 2.5 \times 10^{-3}$ Nm/rad.

With this value, the model closed loop behaviour matched that of the system, as will be shown in the dedicated section.

3.4 Validation of the identified model

First, we analyzed the dynamics of the identified linearized model in the downward equilibrium, whose bode diagram (theta on the left and lambda on the right) is

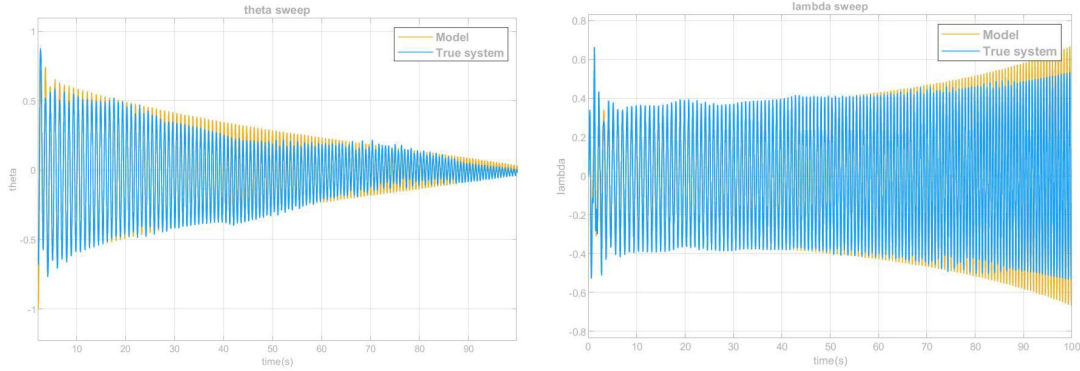


and whose poles are $(-0.4214 \pm 13.0411 i, -0.8095 \pm 2.3451 i)$. Owing to having all negative real parts, the stability of this equilibrium is confirmed.

Then, we collected the response of the system to two different inputs and compared them with the responses of our model:

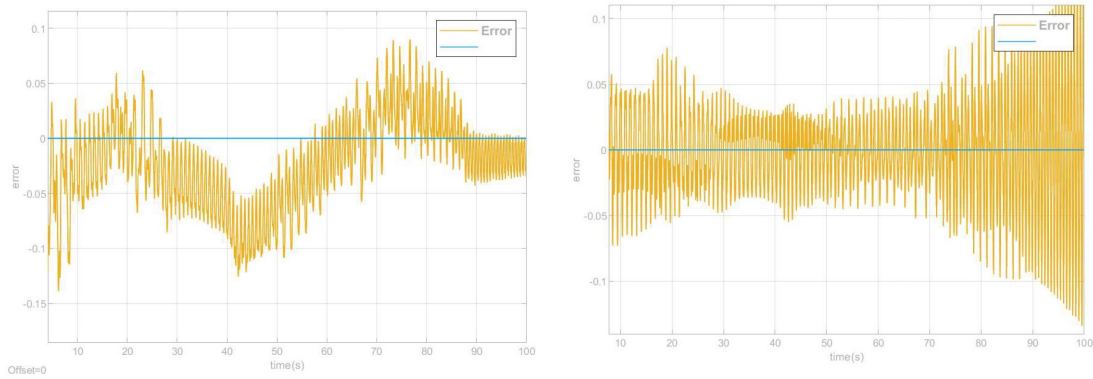
- **Open loop sweep sines**

We fed the system with a 2V sweep sine in the 1-1.7Hz frequency range, selected considering both the physical limits on the horizontal arm angle and range in which the dynamics are revealed, up to the pendulum resonance.



The observed results allowed us to confirm that the model was reflecting the dynamic behaviour of the system, in terms of magnitude and phase within the frequencies of interest.

The following graphs represent the error for theta (left) and lambda (right) between the system and the model



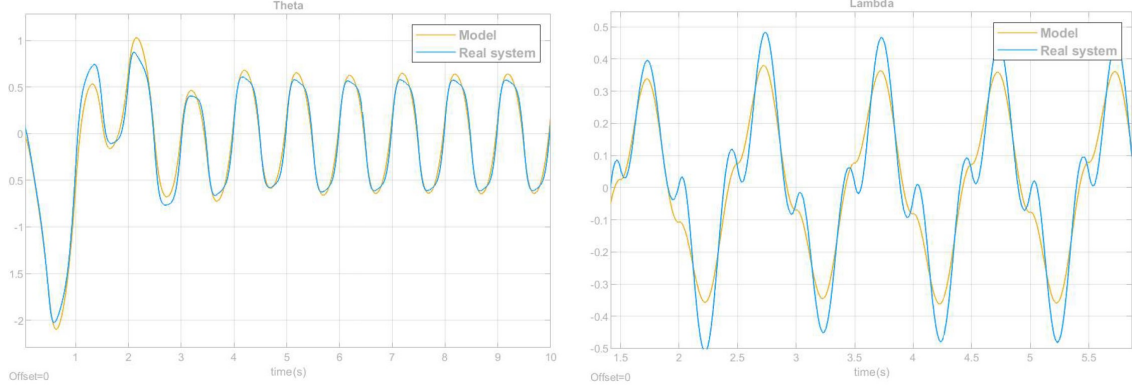
The real system's theta has a low frequency component closely resembling a sinusoidal wave that the model is not catching, which is probably due to unmodeled non-linearities. As the input approaches the resonant frequency of the pendulum the validity of the linearized model ceases due to increased pendulum angle amplitudes.

The RMSEs for all 4 states are:

State	RMSE
θ	0.0837 rad
$\dot{\theta}$	0.3611 rad/s
λ	0.0452 rad
$\dot{\lambda}$	0.6045 rad/s

- **Open loop square wave**

Next, we fed the system with a 1 Hz square wave to observe its behaviour when more harmonics were being excited simultaneously:



The RMSEs for all 4 states are:

State	RMSE
θ	0.0767 rad
$\dot{\theta}$	0.8035 rad/s
λ	0.0711 rad
$\dot{\lambda}$	1.2366 rad/s

The pendulum oscillations in our model are smaller than those of the true system because of the magnitude of the angle being swung stretching the validity of our model, as confirmed by the bigger RMSE. However, overall we were satisfied with the dynamic performances and we proceeded with the development of control strategies.

4 State observer

In order to accurately estimate the states of the plant, it was necessary to correct the estimates made by the model by feeding back the error between the actual and estimated states.

The objective of the filter development was to properly select the feedback gain: the greater it is, the more aggressive the correction will be. However, this approach comes at the cost of signal contamination by the sensor noise.

Therefore, we went on to develop a Kalman filter: for a plant with state equations

$$\begin{cases} \dot{x} = Ax + Bu + Gw \\ y = Cx + Du + v \end{cases}$$

where w represent disturbances and v denotes sensor noise. It is possible to show that the optimal observer feedback gain is given by

$$L = PC^T R_0^{-1}$$

where P is the solution of the Riccati equation

$$AP + PA^T - PC^T R_0^{-1} CP + GQ_0 G^T = 0$$

with Q_0 being the covariance matrix of the disturbances and R_0 being the covariance matrix of the sensor noise.

Accordingly, as the covariance of the sensor is the resolution of the encoder $\frac{2\pi}{2048}$, the R_0 matrix becomes

$$R_0 = \begin{bmatrix} \frac{2\pi}{2048} & 0 \\ 0 & \frac{2\pi}{2048} \end{bmatrix}$$

For what concerns the disturbances, we started using a base value corresponding to the known non-linearities of the system, specifically static friction, by computing the torque corresponding to the static friction voltage (0.4V) and use it as the covariance value in the Q_0 matrix:

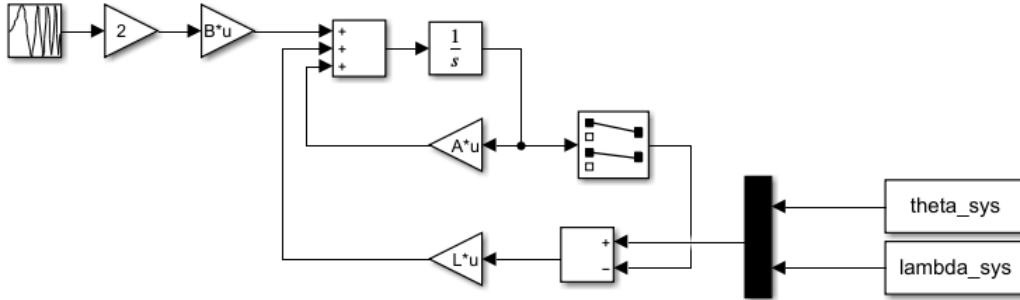
$$T_{staticfriction} = T_{input} = k_t \frac{v(t)}{R_m} = 2 \times 10^{-3} Nm$$

$$\text{therefore } Q_0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 \times 10^{-3} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \times 10^{-3} \end{bmatrix}$$

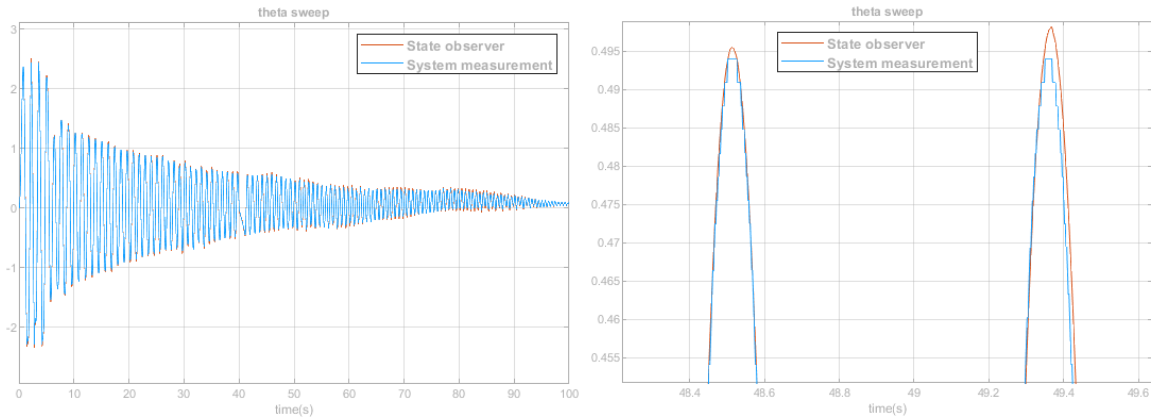
Using the principle of duality and lqr MATLAB function, the corresponding poles introduced by the observer were $(-0.1459 \pm 13.0953i, -0.3166 \pm 2.4631i)$, clearly too slow if compared to both the open and closed loop poles of our system.

As a consequence, we started increasing the variance related to the disturbances, which was very likely to be much higher than the static friction torque alone as testified by the RMSE of the two speed states, until we obtained a gain matrix that was high enough to yield observer poles that were ≈ 10 times faster than the slowest closed loop ones. The final poles of the observer were $(-29.9666 \pm 30.1239i, -25.8596 \pm 22.3502i)$.

Implementing the observer depicted in the following scheme



and testing it with the same sweep signals we validated the model on, we obtained the following result:



the state observer is not influencing the relevant system dynamics while also filtering out the discrete steps of the encoder, as visible in the closeup.

5 Control of the horizontal arm

5.1 PID control

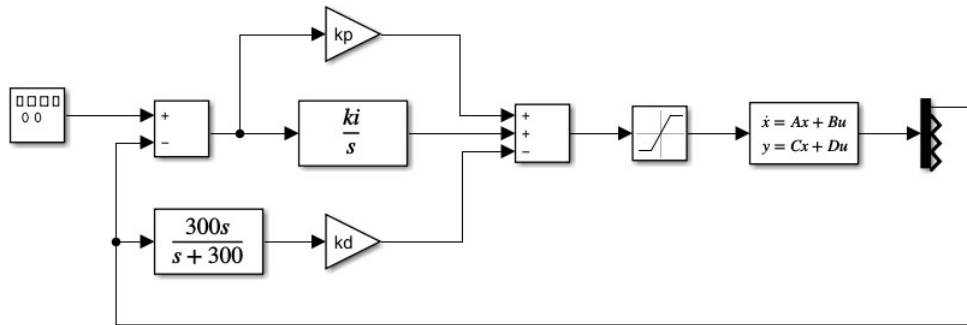
When facing the challenge of developing a PID controller for a 4th order system, our initial approach was to attempt to approximate it as a second order system so that we could obtain an analytical solution starting from self given specs. However, because the slowest poles are just half as fast as the fastest ones, we couldn't really consider them as dominant.

Subsequently, we discarded that approach and attempted to design the PID controller using classical techniques such as root locus, to achieve desired performances; with the root locus method we were able to drive some of the poles to desired location. However, it wasn't an effective technique to drive all the poles.

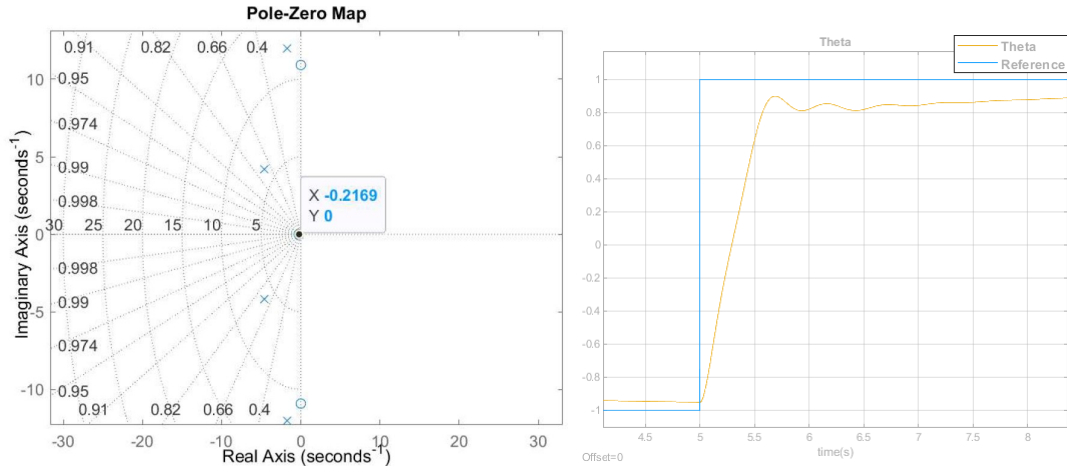
Ultimately, our final approach was to start from a base tune for the PID controller parameters and, considering the dynamics of the system, fine tune the controller to ensure that:

- The cutoff frequency was low enough to properly attenuate the pendulum resonance.
- The system had small to no overshoot.
- The control action wasn't being saturated.
- The system had the smallest possible settling time.
- Steady state oscillations were minimized.

With our goal in mind, we implemented the "real" PID scheme, where the derivative action is performed on the feedback and not on the error, using also a low pass filter, as depicted in the following scheme



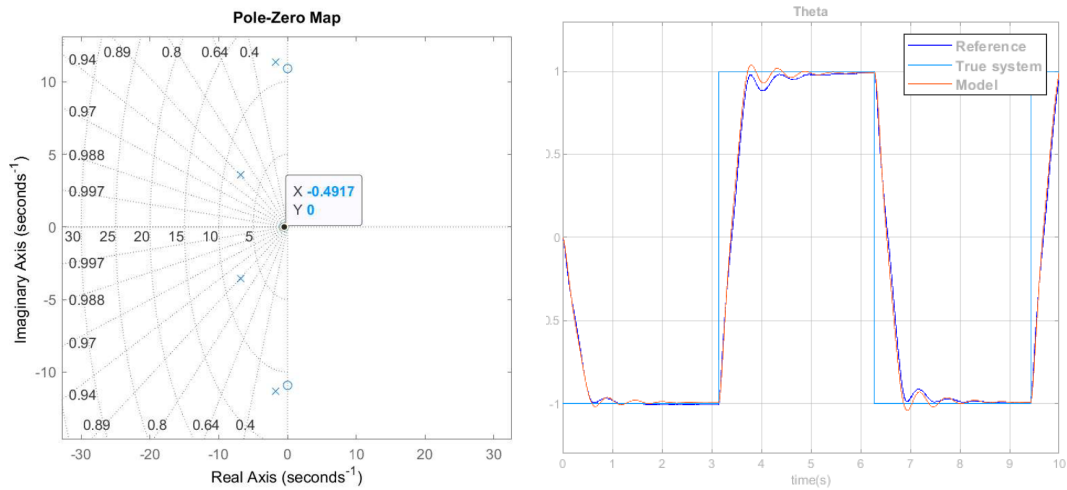
At this point, starting from low initial values ($k_p = 2, k_d = 0.5, k_i = 0.5$), we analysed the closed loop dynamics by referencing to the bode plot and looking at the position of the poles:



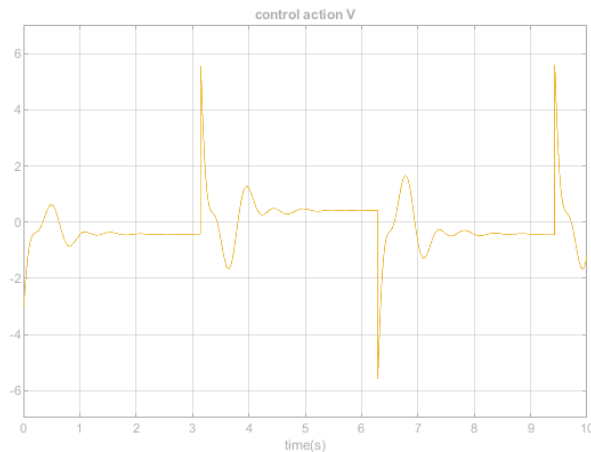
With these parameters, the system behaves like a second order underdamped system, as evidenced by the pole-zero map, whose dominant closed loop poles are at $(-4.63 \pm 4.2i)$ corresponding to a cutoff frequency of $\approx 4.4 \text{ rad/s}$, with an actuator input of approximately 4v. The system response shows a very slow steady state error correction, hinting that a higher integral action is needed.

Holding on to the knowledge gained from this initial analysis, we went on with the tuning of the parameters. A higher integral action and a higher gain were implemented in order to have a faster response, without going too close to the resonance with the cutoff frequency.

This process led us to the final tuning parameters ($k_p = 3, k_d = 0.7, k_i = 1.5$):



which performed the best, even though the corresponding control action is lower than other strategies we tried and far from the saturation limit as it can be seen here:



The system closed loop poles are located at $10^2 \times (-2.8945, -0.0213 \pm 0.1200i, -0.0414 \pm 0.0599i, -0.0047)$, with the dominant ones being $(-4.14 \pm 5.99i)$. Therefore, the system behaves like a second order underdamped system, whose cutoff frequency is at 7.15 rad/s, with a damping ratio of ≈ 0.6 and percentage overshoot of 10%.

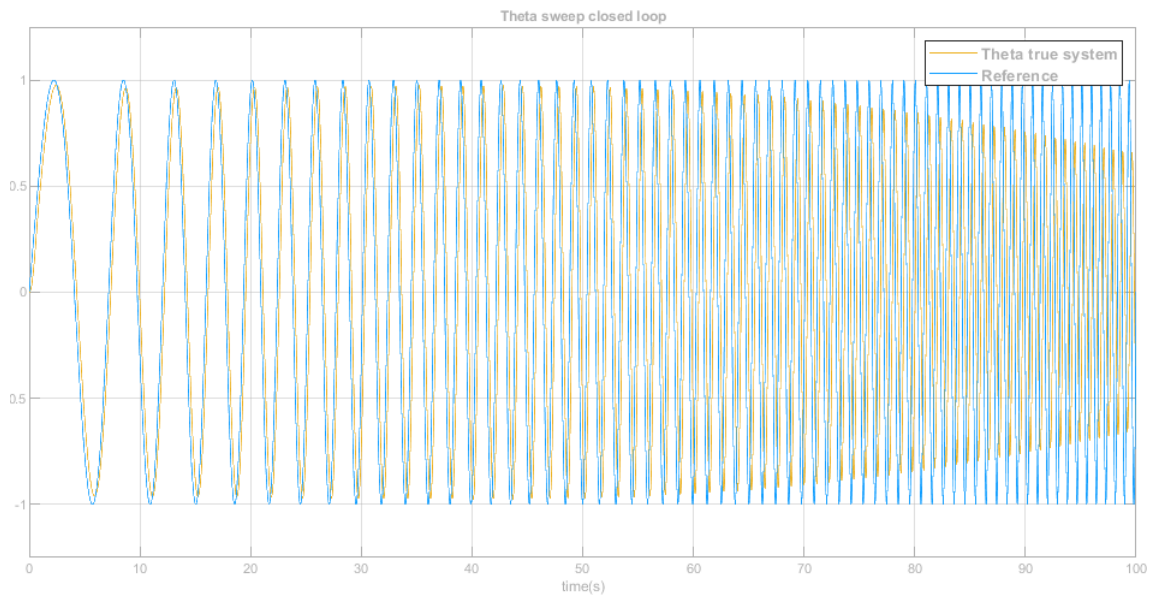
- **Performance analysis of the final PID controller**

By analyzing the response of the closed loop system, the key performance indexes are reported as follows:

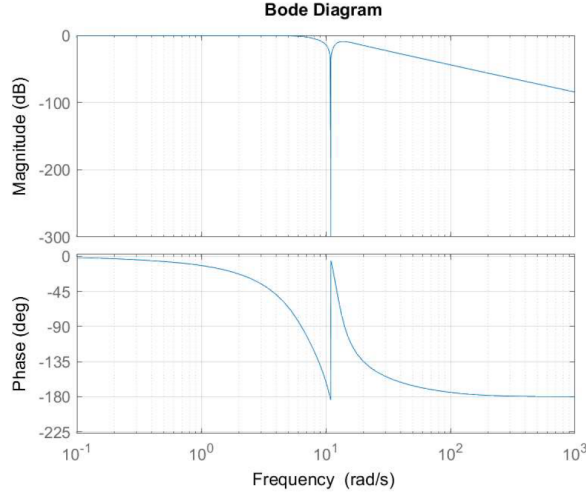
Performance indicator	Value
Rise time(10%-90%)	0.456 s
Settling time(5%)	1.01 s
Percentage overshoot	4 %
Steady state error	0 rad (integrator)
Phase margin	180 deg
Gain margin	68.9 dB @ 10.9 rad/s

It's worth noting that the true system overshoot was lower than that of the model, probably because of the neglected static friction effect.

Next, we tested the bandwidth of the closed loop by feeding a sweep sine as a reference in the 0.1-1.2 Hz frequency range:



We observed a -3db attenuation at around 1.13 Hz, in line with the expected behaviour deduced from the bode plot of the model:



Ultimately, we were satisfied with the archived performances. Opting for this "cheaper" control strategy instead of more expensive ones in terms of actuator usage increased the rise time by approximately 5%; however, it decreased the steady state oscillation and settling time owing to the lower bandwidth.

5.2 State space control for the horizontal arm

In order to see if we could further reduce the oscillations induced by the pendulum and achieve better performances with respect to that achieved by the PID controller, we implemented more advanced control techniques:

- **LQR control**

To fully exploit optimal control, we needed to properly select the parameters in the LQ cost function

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

where Q and R are square, symmetric, positive definite matrices. Accordingly, we started with initial values based on Byron's rule:

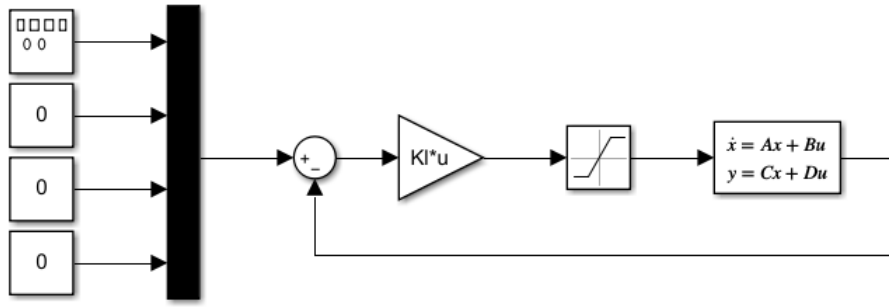
$$Q_{ii} = \frac{1}{\text{maximum acceptable value for } x_i^2} \quad \text{with } i = 1, \dots, n$$

$$R_{ii} = \frac{1}{\text{maximum acceptable value for } u_i^2} \quad \text{with } i = 1, \dots, n$$

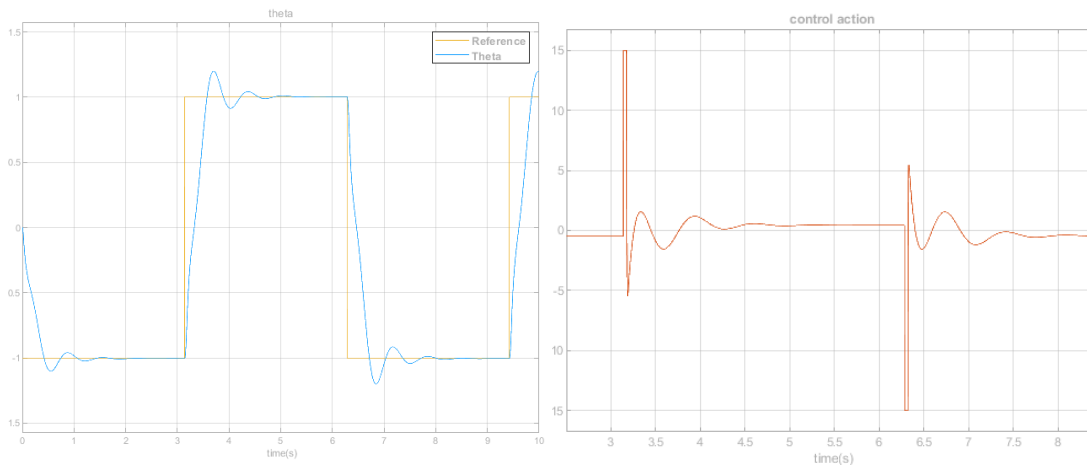
where, imposing a maximum deviation for theta of 0.1 rad and for u of 15 V according to our goal of penalizing oscillations while not saturating led to

$$Q = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R = 4.4 \times 10^{-3}$$

with the other entries are left as one. Implementing the controller on the model as in this scheme



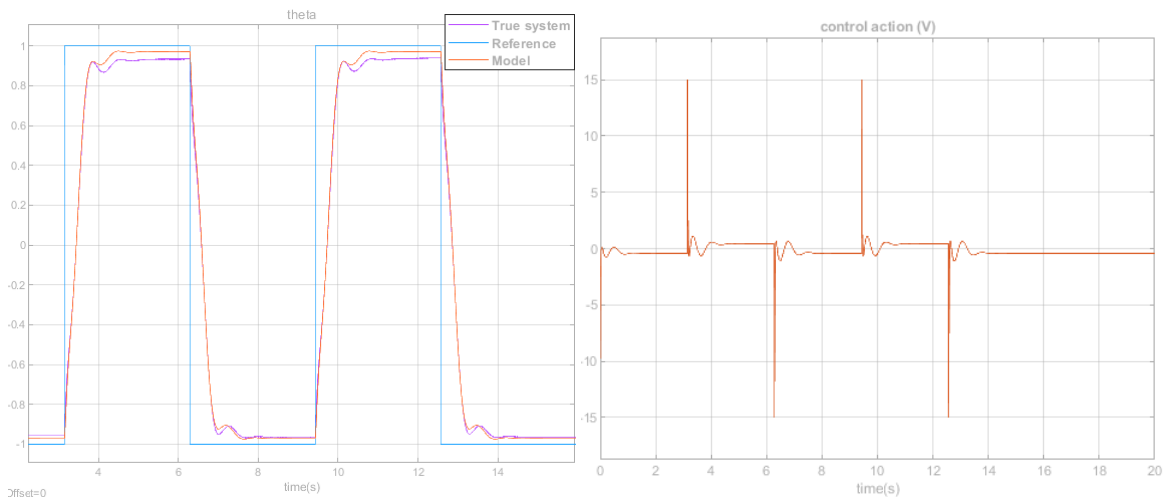
where K_l is the gain matrix obtained with MATLAB's `lqr` function, resulted in the following system behaviour:



whose poles are located at $10^2 \times (-4.0927 + 0.0000i, -0.0937 + 0.0000i, -0.0257 \pm 0.0938i)$. From this test it was quite evident that the control action was saturating, with the system overshooting by 10%. Following these considerations, we tuned the two matrices following these steps:

- increasing the R value until the system didn't saturate
- decreasing the θ entry in Q until we observed very small to no overshoot, while also observing the changes in the dynamic behaviour of the system

Finally, we were able to archive the best compromise with $Q_{11} = 20$ and $R = 0.2$; implementing it onto the real system yielded the following response:

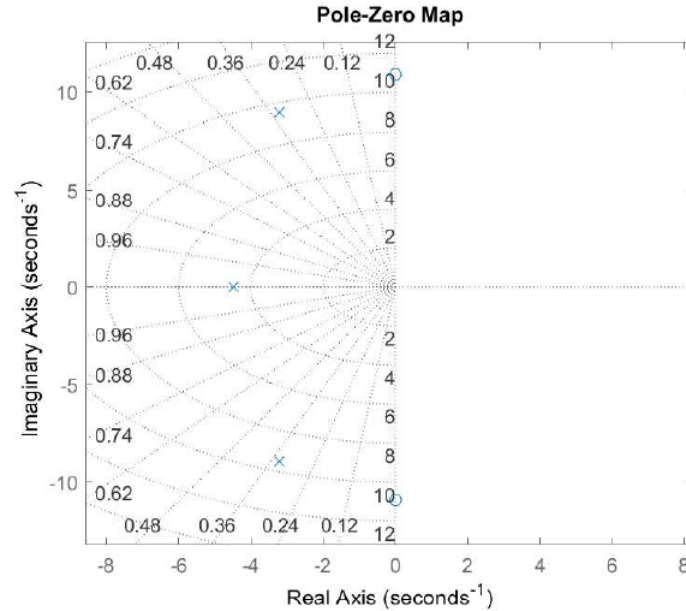


whose closed loop poles are located at $(-68.4933, -4.4883, -3.2106 \pm 8.9504i)$, resulting in a cutoff

frequency of 7.2 rad/s.

By looking at the difference between the model and system responses, an asymmetric behaviour was noticed. When the wave was positive, the system wasn't reaching the same steady state value of the model, but the same wasn't true when the wave was negative; we attributed this difference to the non symmetric (therefore non linear) torsional stiffness of the cable.

The system behaves almost like a first order one because of the real dominant pole (even though there is a slight influence from the pendulum oscillation), as can be seen from the pole-zero map



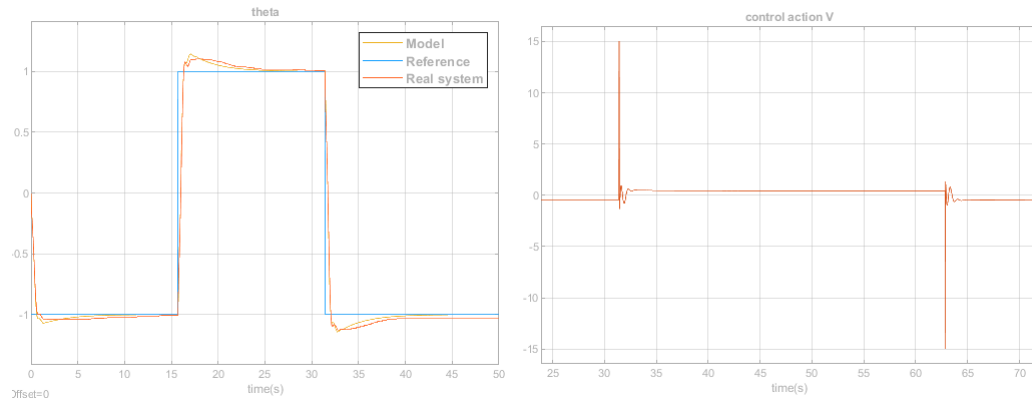
where the introduced zeros are close to the complex poles. Overall, the key performance indicators are:

Performance indicator	Value
Rise time(10%-90%)	0.558 s
Settling time(5%)	1.1 s
Percentage overshoot	no overshoot
Steady state error	0.07 rad
Phase margin	180 deg
Gain margin	68.9 dB @ 10.9 rad/s

• LQR control with integral action

In order to eliminate the steady state error, we tried to implement an integral action in the LQR controller by augmenting the state vector of the system with the integral of the error. As a result, we were able to penalize the steady state error, even though the influence of the parameter on the closed loop dynamics wasn't really predictable during the tuning process.

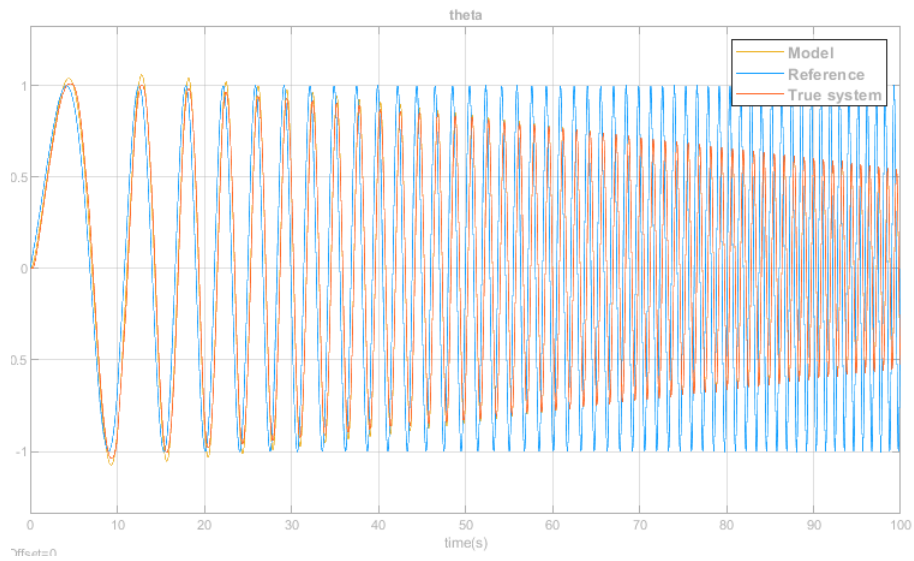
As it can be seen here



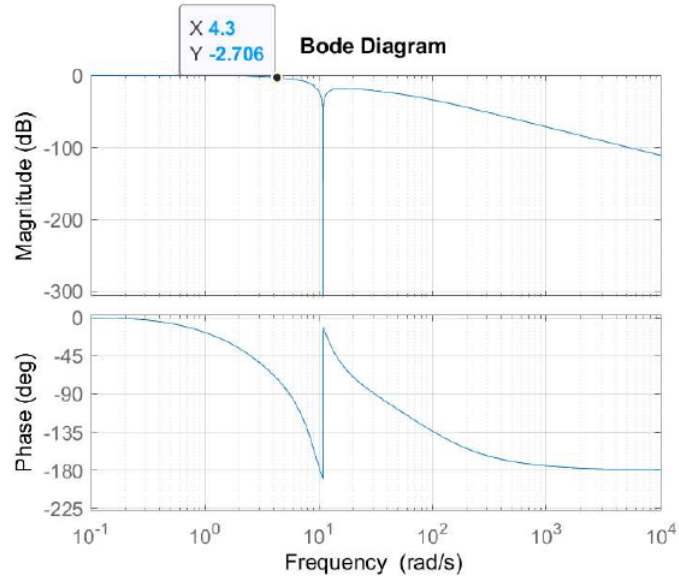
the system response was quite similar to the one without integration, and the steady state error compensation was considerably slow, making this development not really effective. Looking at the dynamic behaviour, the poles of the closed loop are located at $(-84.5397, -4.3517, -3.1955 \pm 8.9213j)$, the dominant one being -4.3517 . The key performance indicators are:

Performance indicator	Value
Rise time(10%-90%)	0.46 s
Settling time(5%)	4.37 s
Percentage overshoot	7%
Steady state error	0 (integrator)

In order to verify the bandwidth of the controller, we fed a sweep sine as a reference in the 1-1.7Hz frequency range



Comparing it to the bode plot of the closed loop



allowed us to confirm a cutoff frequency of 4.52 rad/s.

5.3 Comparison between proposed control strategies

Overall, between the strategies we examined, the pros and cons of each one are reported as follows:

- **PID**

Pros	Cons
Lowest control effort	Overshoots
Fastest overall, both in rise and settling time	Shows the most oscillatory behaviour
Has a relatively high bandwidth	-
Not influenced by the non-linearity of the cable	-

- **LQR**

Pros	Cons
Closest behaviour to a first order system	Steady state error
No overshoot	Not the fastest
-	Suffers the cable non linearity

- **LQR + integral action**

Pros	Cons
Fixes the steady state, but very slowly	Slowest overall
Doesn't suffer cable non-linearity	Biggest overshoot of the three

Taking into account these considerations, the best control strategy for the horizontal arm in the lower equilibrium is the PID controller that archives the fastest dynamic response while being the cheapest control strategy, with a required actuation effort of 6V.

6 Balance control in the unstable equilibrium

At this point, we proceeded with the development of control strategies for the upward equilibrium, with the ultimate goal of stabilizing the pendulum (manually brought up), while also following a reference on the horizontal arm angle.

Analyzing the linearized model of the system, whose poles are $(-13.3024, 12.5508, -0.8552 \pm 2.3548i)$, the equilibrium instability is evident, given the presence of

a positive real pole.

To solve this problem, we resorted to state space methods, starting out with LQR, because it could give us more insights on the effects introduced by the dominant pole position. Subsequently, we exploited the acquired knowledge to manually place the poles using pole placement.

6.1 LQR control

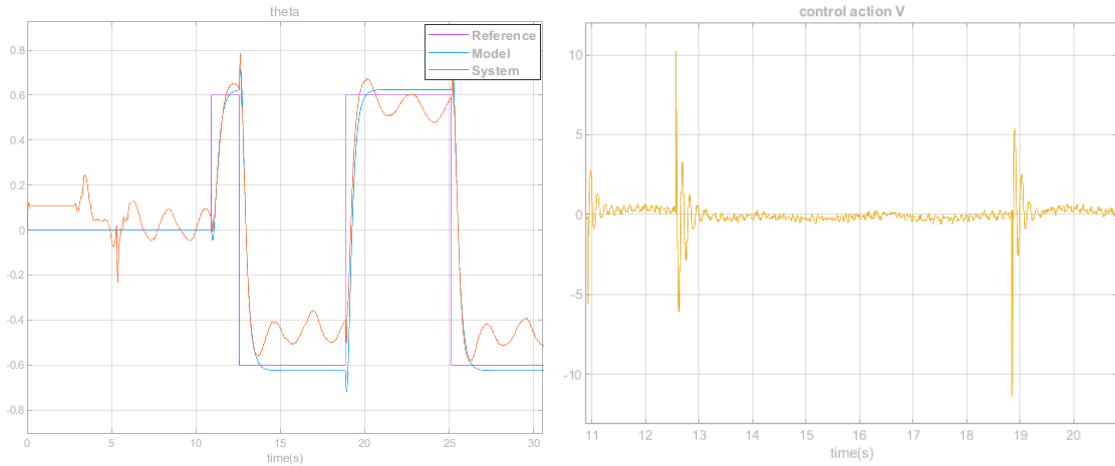
For the the cost function weights in the upward case, we penalized both the horizontal and pendulum angles, trying to observe how changing these parameters affected the system dynamics, while keeping an eye on the saturation limit for the control variable.

To implement the controller, we inserted a block in the scheme that made the pendulum angle equal to 0 when upward and a logic switch that enabled the controller only within 10 degrees from the vertical position. Accordingly, we were able to manually bring the pendulum up with the interface already running.

After some trial and error, we ended up with the following Q and R matrices:

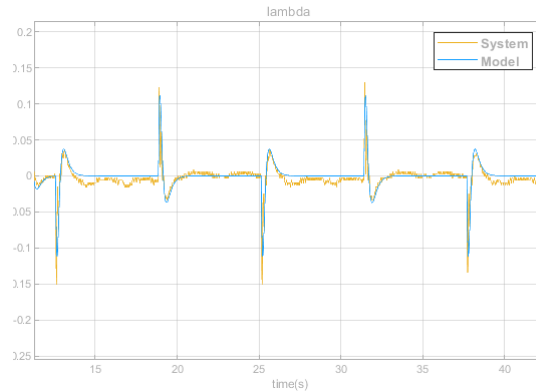
$$Q = \begin{bmatrix} 15 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 20 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R = 0.1$$

where the most penalized angle in the matrix is the pendulum one



The closed loop poles corresponding to these parameters were $(-86.94, -8.7526 \pm 3.1788i, -3.9106)$, confirming the stability of the controlled system.

It's quite evident from the system response to step references that there is a small steady state oscillation that the model is not catching: we interpreted it as the balancing action that the system needs to perform in order to keep the pendulum in the equilibrium in the presence of non linearities, especially static friction. Looking at the pendulum angle



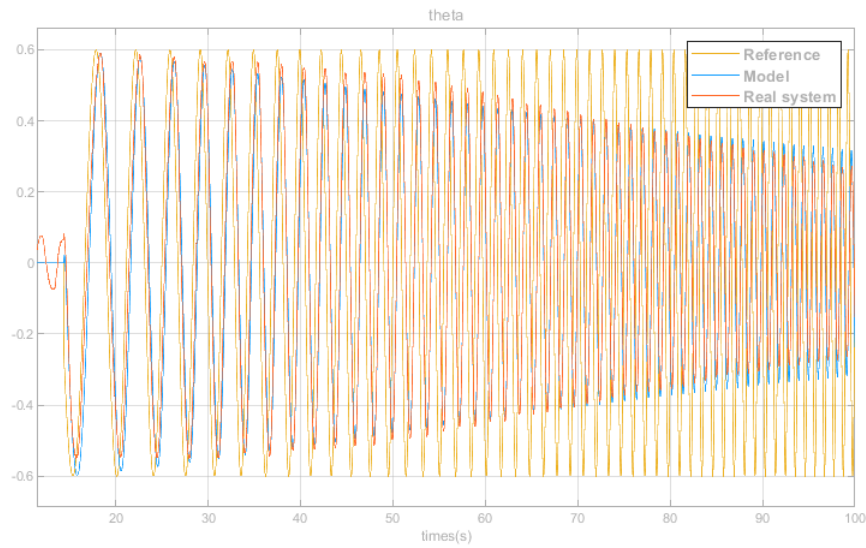
confirms our intuition because the pendulum is oscillating around the equilibrium at the same frequency of the arm oscillations.

- **Controller performances**

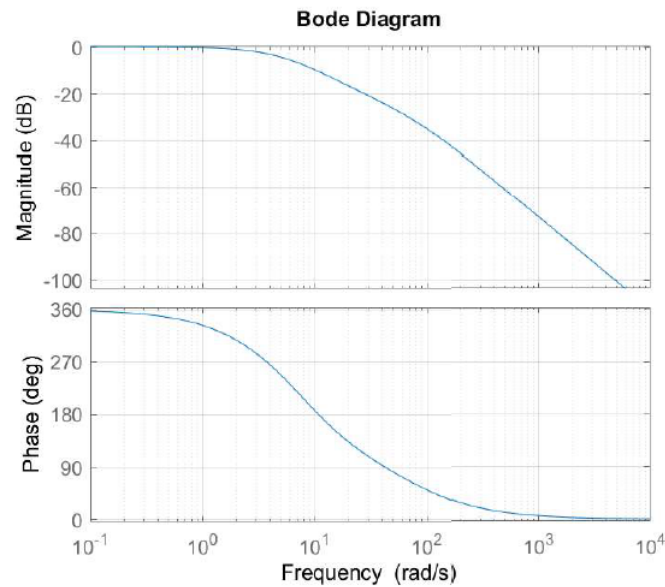
Overall, the key performances indicators are:

Performance indicator	Value
Rise time(10%-90%)	0.53 s
Settling time(5%)	never settles s
Percentage overshoot	15%
Steady state error	depends on direction
Gain margin	9.86 dB @ 11.3 rad/s
Phase margin	130 deg

Sending a sweep sine reference to verify the bandwidth



and cross-referencing the results with the bode plot of the closed loop system



confirms a cutoff frequency of 4.5 rad/s. From the sweep response, notice that, as in the lower equilibrium case, the closed loop is not able to compensate for the asymmetric behaviour of the cable, which exhibits a stiffer behaviour along one of the rotation directions.

We also noted this behaviour changes based on how the cable was plugged in, adding more uncertainty into the system.

6.2 Pole placement control

Next we went on with the development of a pole placement controller, to assess whether by manually placing the poles we could obtain a better response from the system.

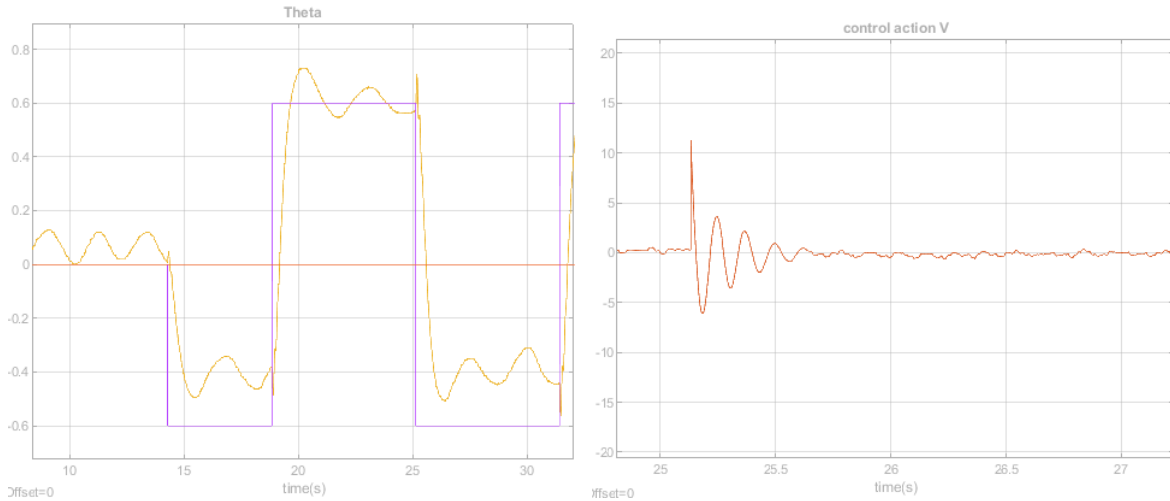
Exploiting the results of the lqr, we set the reference dominant pole as (-3.91):

- Our first approach was to try to implement a faster controller, by placing the poles at around -4.5 rad/s. However, even though the controller was able to maintain the pendulum upright, it wasn't stable when fed with a square reference owing to having too low stability margins.
- Keeping the above result in mind, we set the real part of the dominant complex pair equal to the lqr one, while the two remaining poles were placed as far as possible, making sure that the feedback gains weren't too high.

With the desired poles being

$$p = [-3.9 \pm i, -30, -40]$$

the system response was:



In this case the behaviour is very similar to the lqr controlled system, with an even more pronounced influence from the encoder cable, as it can be observed by the average arm angle when the reference is 0. It's difficult to tell if this difference is due only to the controller because the experiments were carried out in two different sessions, meaning that the cable could have been plugged in with a different orientation, yielding a slightly different torsional stiffness.

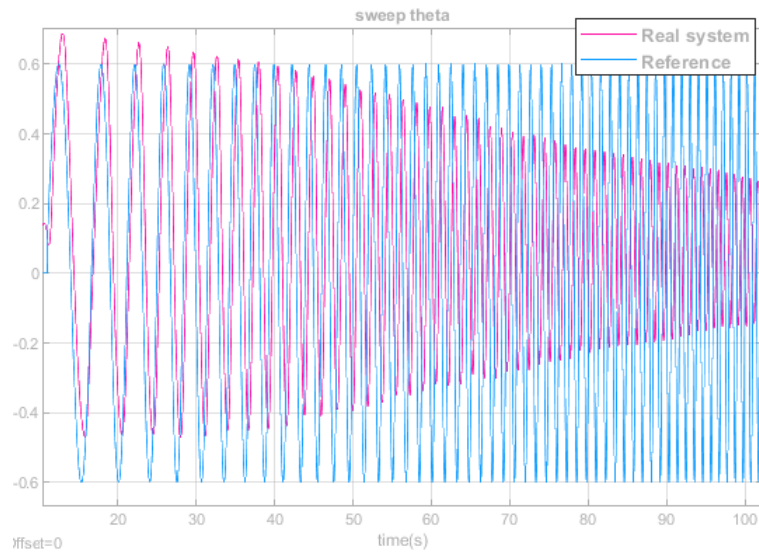
In terms of actuator usage, the pp and lqr strategies were very similar, both peaking at approximately 10 V when the reference changed signs.

• Controller performances

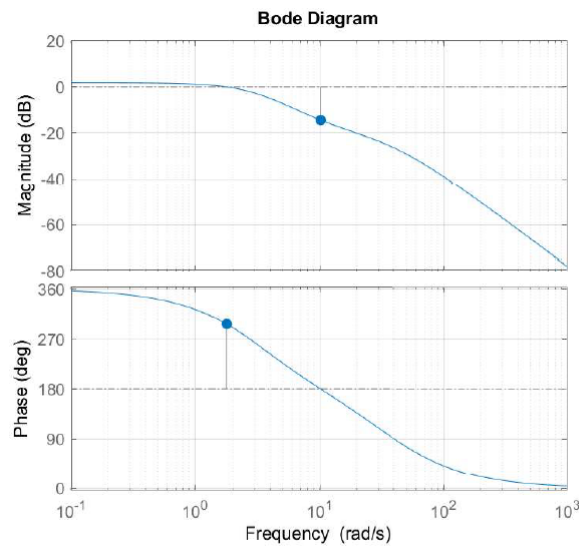
Overall, the key performance indicators are:

Performance indicator	Value
a Rise time(10%-90%)	0.53 s
Settling time(5%)	never settles s
Percentage overshoot	20%
Steady state error	depends on direction
Gain margin	12.4 dB @ 11.6 rad/s
Phase margin	126 deg

Then, we fed the system with a sweep sine

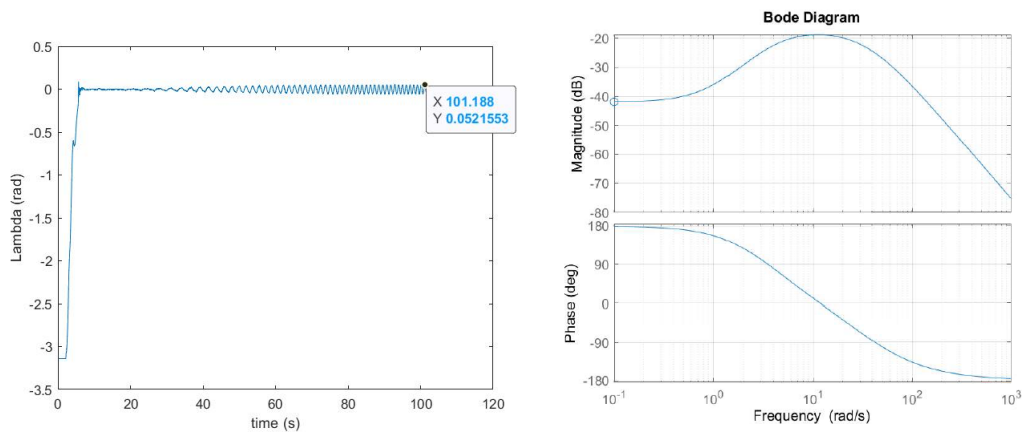


and compared the results to the bode plot of the closed loop system to verify the controller bandwidth



The analysis confirmed a bandwidth of 3.72 rad/s.

Confronting also the sweep response of the pendulum angle with the respective bode plot



confirmed the behaviour we observed: a good attenuation of oscillations up until the resonance, where they were nevertheless contained within peaks of ≈ 0.06 rad.

6.3 Comparison between the two strategies and final balance control implementation

The two controllers exhibited considerably similar behaviours, as one could expect from the placement of the poles. Our choice for the final balance controller to be implemented in the swing up scheme was the pole placement controller, because of the degree of freedom we had when assigning the dominant poles.

This choice revealed itself to be useful as soon as we started swinging up the pendulum. With the already selected poles, the system was stable when brought up manually; however, it wasn't when it had to catch the pendulum arriving with a more significant amount of energy.

In order to solve this issue, we moved the two dominant poles from -3.9 to -3 rad/s, resulting in the following characteristics:

Performance indicator	Value
a Rise time(10%-90%)	0.6 s
Settling time(5%)	never settles s
Percentage overshoot	20%
Steady state error	depends on direction
Gain margin	14.4 dB @ 10.2 rad/s
Phase margin	117 deg

with a bandwidth of 3.1 rad/s.

The higher gain margin of this strategy allowed to stabilize the system even when being swung up, confirming its increased robustness at the cost of a lower bandwidth; therefore, it was not able to track faster signals.

7 Swing up control

The final task was to make the pendulum swing up from the downward position and balance it in the unstable equilibrium. For this purpose, we split the control problem in two: the second part, namely, balancing the pendulum, was already solved with our final pole placement controller. For the swing up, we tried two different approaches, with only one being consistently successful.

7.1 Resonance excitation

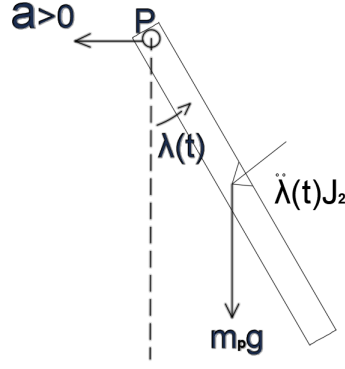
We tried to excite the pendulum using a square wave characterized by the pendulum resonant frequency, so that oscillations could be amplified until they were big enough for the balance controller to kick in. This approach revealed itself to be successful only half of the time, with an inconsistent number of preparatory oscillations.

We also tried to vary the amplitude of the input wave, making it larger at the beginning, which was also unsuccessful and led us to discard this strategy altogether.

7.2 Energy based swing up

The second approach was a non linear energy-based control strategy, whose core idea is the preservation of the total energy in ideal (neglecting friction) systems.

Starting from the ideal pendulum



where $a(t)$ is defined as the linear acceleration of the pivot point P, positive accordingly to our conventions, we proceeded to write the equation of motion along the direction perpendicular to the pendulum shaft

$$J_p \ddot{\lambda}(t) + m_p g \frac{L_p}{2} \sin \lambda(t) - m_p \frac{L_p}{2} a(t) \cos \lambda(t) = 0$$

and define the kinetic and potential energies of the pendulum

$$K(t) = \frac{1}{2} J_{p_{cm}} \dot{\lambda}(t)^2 \quad , \quad V(t) = m_p g \frac{L_p}{2} (1 - \cos \lambda(t)).$$

Defining now the total energy of the system and its derivative

$$E(t) = V(t) + K(t) = \frac{1}{2} J_{p_{cm}} \dot{\lambda}(t)^2 + m_p g \frac{L_p}{2} (1 - \cos \lambda(t))$$

$$\dot{E}(t) = J_{p_{cm}} \ddot{\lambda}(t) \dot{\lambda}(t) + m_p g \frac{L_p}{2} \sin \lambda(t) \dot{\lambda}(t)$$

and substituting $J_p \ddot{\lambda}(t)$ with the motion eq., we were able to rewrite the derivative of the total energy as

$$\dot{E}(t) = m_p \frac{L_p}{2} a(t) \cos \lambda(t) \dot{\lambda}(t).$$

This relationship highlights the proportionality between the derivative of the total energy and linear acceleration of the pivot point, which is also proportional to the voltage being applied at the motor:

$$m_{arm} a(t) L_{arm} = T_{in} = k_t \frac{v(t)}{R_m} \implies v(t) = \frac{R_m}{k_t} L_{arm} m_{arm} a(t).$$

Defining now the Lyapunov function

$$V(t) = (E_r - E(t))^2 / 2$$

where E_r is the reference energy, defined as the potential energy of the pendulum when upward, and taking its derivative with respect to time

$$\dot{V}(t) = -m_p \frac{L_p}{2} K((E_r - E(t)) \cos \lambda(t) \dot{\lambda}(t))^2$$

it's quite immediate to notice that it's always negative, expect for two cases:

- $\dot{\lambda}(t) = 0$
happening at the beginning and (ideally) at the upward position
- $\cos \lambda(t) = 0$
happening at $\frac{\pi}{2}$, where, however, the system cannot maintain its position

Therefore, we employed the following proportional law as the control strategy

$$u = sat_{umax}(K(E_r - E(t)) sign(\cos \lambda(t) \dot{\lambda}(t)))$$

to drive the system to $E(t) = E_r$, which requires an initial "kick" to make $\dot{\lambda}(0) \neq 0$.

In order to implement this strategy, we needed to tune two main parameters: the saturation limits

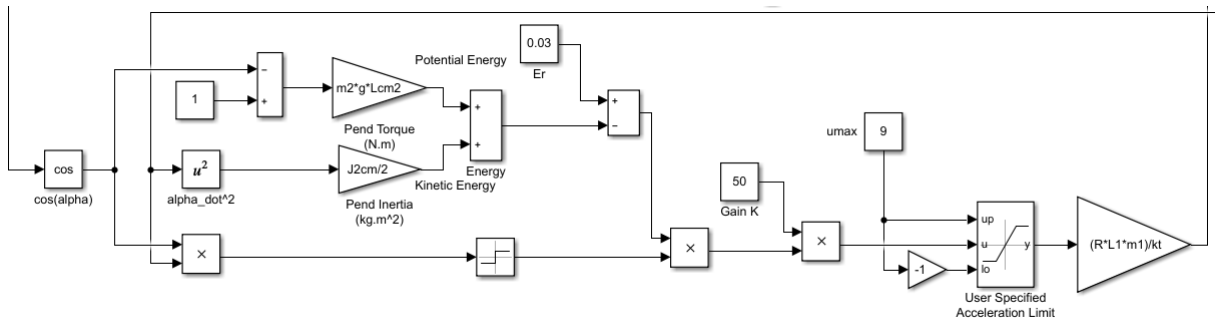
for the linear acceleration and proportional gain K.

For what concerns the first, we considered the above stated relationship between voltage and linear acceleration, imposing the saturation limit of the actuator and obtaining

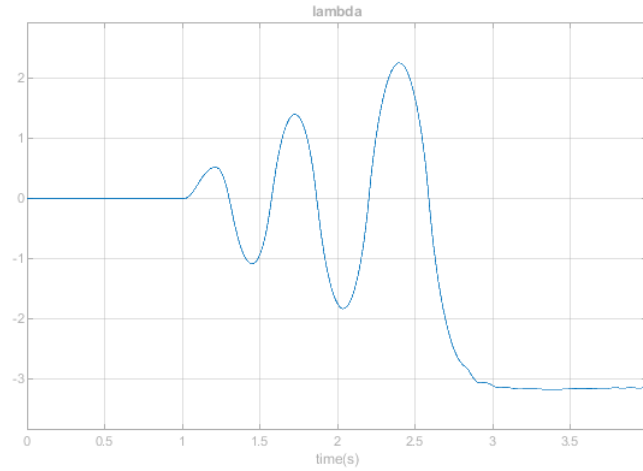
$$a_{max} = 9.28 \frac{m}{s^2}.$$

The tuning of the gain wasn't as straightforward, not having implemented a full non linear model. Therefore, we tried the control law with the model linearized in the downward equilibrium and checked the entity of the control action corresponding to the very first swings, ultimately selecting a conservative value for the starting gain.

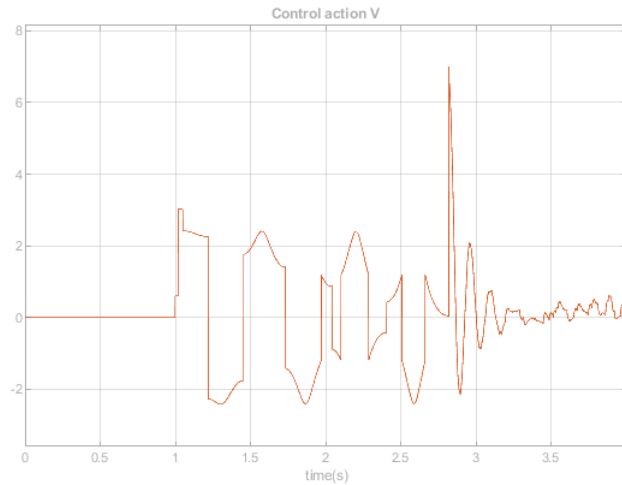
Finally, implementing the controller reported in the following scheme and converting the pendulum angle to be 0 when upward



with the proportional gain set to 50, led to the swing up of the pendulum in three full oscillations.



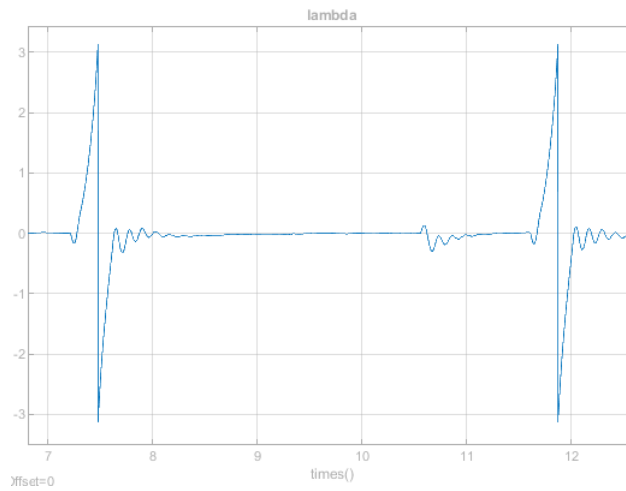
The corresponding control action is



We tried to implement bigger initial pulses, to see whether we could speed up the energy build up process; however, doing so yielded two undesired effects:

- If the pulse amplitude exceeded the initial control action of the Lyapunov controller, the overall number of oscillations increased. This was probably because we were moving the system further from the Lyapunov surface, meaning that when the pulse was over the controller had to bring it back, adding additional swings.
- The additional energy wasn't enough to reduce the number of oscillations, instead it increased the pendulum speed (and therefore the energy) when reaching the balance control handover window. Because of this, the balance controller wasn't able to stabilize the pendulum, effectively saturating the control action; thus, the pendulum kept on cycling.

After the tuning process, we tested the the swing up controller when the pendulum was being manually disturbed, to the point where the angle was out of the balance controller domain, to see if it was able to bring it up faster since the pendulum already had stored potential energy



The controller was always able to bring it up in a small amount of oscillations; in certain cases it in needed just one.

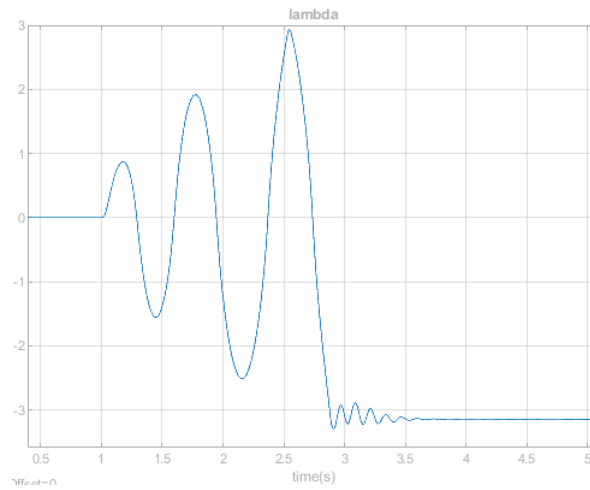
7.3 Additional Lyapunov based approach

Another approach we tried was a control law proportional to the exponential of the pendulum angle:

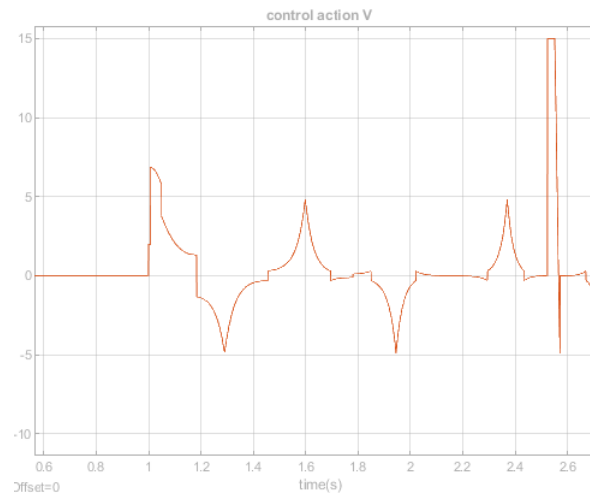
$$u = \text{sat}_{u_{max}} K |\lambda(t)|^n \text{sign}(\dot{\lambda}(t) \cos \lambda(t))$$

the idea here was to tune the exponent so that the control action was much higher when downward, exponentially decaying as the angle approached the upward equilibrium, to see if we could achieve a smoother transition.

This strategy revealed itself to be considerable tricky to tune. Moreover, we were able to swing up the pendulum with the same amount of oscillation as the other strategy



but the control action had higher peaks; in addition, the energy with which the pendulum reached the equilibrium made the balance controller saturate



Overall, the best strategy was the energy based method, demonstrating a swing up time of 2.8s and three full oscillations. The second proposed strategy, although requiring only 2.5 seconds to swing up, made the balance controller saturate, resulting in a rough handover between the controllers and causing the pendulum to oscillate.